

# Equalizer Lowpass FIR Filter Design via Convex Optimisation

## 1 Overview

A complex-valued lowpass FIR filter is designed whose passband target matches the UCDC equalizer frequency response. The filter is found by solving a second-order cone programme (SOCP) using CVXPY.

### Signal chain parameters

| Parameter                     | Value                                        |
|-------------------------------|----------------------------------------------|
| Sample rate $f_s$             | 750 MHz (two-sided, Nyquist = $\pm 375$ MHz) |
| Passband edge $f_p$           | $\pm 162.5$ MHz                              |
| Stopband edge $f_{stop}$      | $\pm 250.0$ MHz                              |
| Filter taps $N$               | 63                                           |
| Passband tolerance $\delta_p$ | 0.05                                         |
| Stopband tolerance $\delta_s$ | 0.1                                          |

## 2 Optimisation Formulation

The DTFT of the complex tap vector  $\mathbf{h} \in \mathbf{C}^N$  is

$$H(\omega) = \sum_{n=0}^{N-1} h[n] e^{-j\omega n} = \mathbf{a}(\omega)^\top \mathbf{h}, \quad \mathbf{a}(\omega) = [1, e^{-j\omega}, \dots, e^{-j\omega(N-1)}]^\top.$$

Sampling over a frequency grid turns this into the linear map  $\mathbf{H} = \mathbf{A}\mathbf{h}$ , where  $A_{kn} = e^{-j\omega_k n}$ .

Let  $\mathcal{P}$ ,  $\mathcal{S}$ , and  $\mathcal{T}$  denote the index sets of the passband, stopband, and transition-band frequency grid points, respectively. The **fixed-tolerance** problem is

$$\begin{aligned} & \underset{\mathbf{h}}{\text{minimise}} && \sum_{k \in \mathcal{P}} |\mathbf{a}(\omega_k)^\top \mathbf{h} - H_d(\omega_k)| + \lambda \|\mathbf{h}\|^2 \\ & \text{subject to} && |\mathbf{a}(\omega_k)^\top \mathbf{h} - H_d(\omega_k)| \leq \delta_p, \quad k \in \mathcal{P} \\ & && |\mathbf{a}(\omega_k)^\top \mathbf{h}| \leq \delta_s, \quad k \in \mathcal{S} \\ & && |\mathbf{a}(\omega_k)^\top \mathbf{h}| \leq \delta_{t,k}, \quad k \in \mathcal{T}. \end{aligned}$$

The transition-band bound  $\delta_{t,k}$  decreases linearly from  $\max_{k \in \mathcal{P}} |H_d(\omega_k)|$  at the passband edge to  $3\delta_s$  at the stopband edge, preventing large overshoot in the unconstrained gap while keeping the SOCP feasible.

## 3 Filter Response

Figure 1 shows the designed filter's magnitude (dB) and phase response overlaid with the desired equalizer response  $H_d(\omega)$ .

Figure 2 shows the in-band magnitude response zoomed to  $\pm 5$  MHz beyond the passband edges, with the y-axis limited to  $[-0.9, +0.1]$  dB to resolve the passband ripple.

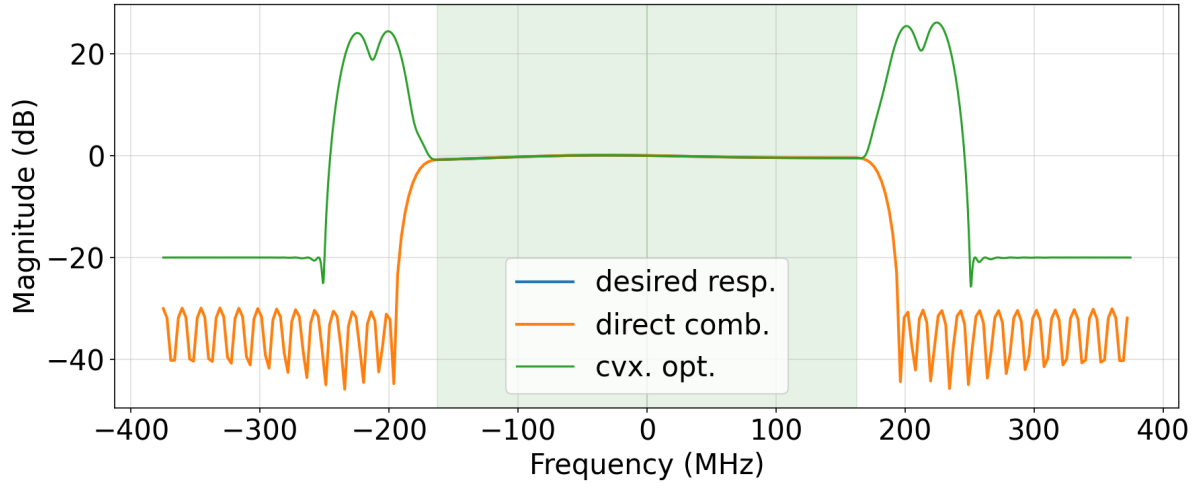


Figure 1: Magnitude (top) and phase (bottom) of the designed equalizer lowpass FIR filter. The green shaded regions indicate the passband  $\pm 162.5$  MHz. The dashed red line is the desired response  $H_d(\omega)$  interpolated from the 256-bin UCDC equalizer FFT.

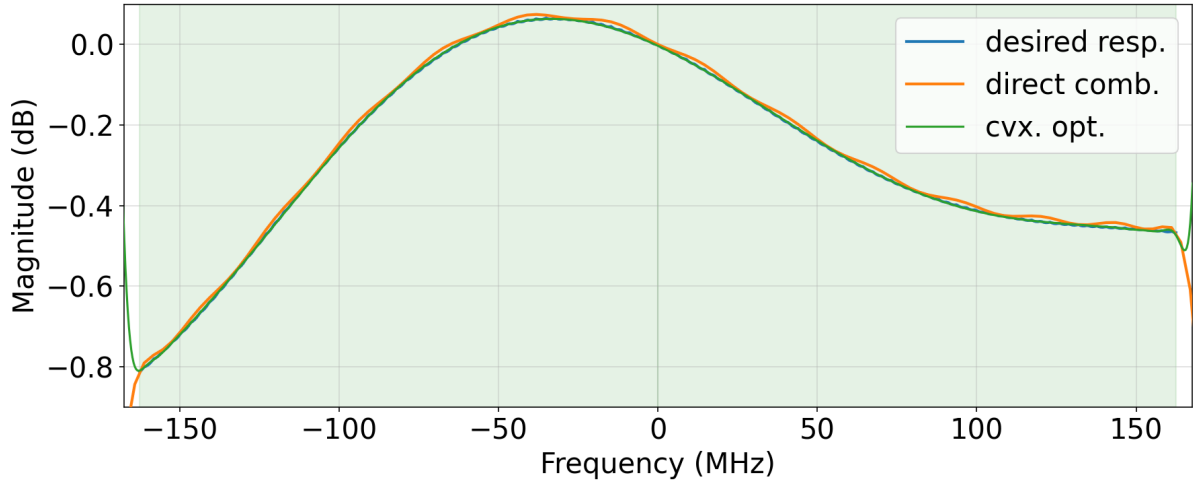


Figure 2: In-band magnitude response zoomed to  $[-167.5, +167.5]$  MHz with a  $[-0.9, +0.1]$  dB y-axis. The green shaded regions indicate the passband  $\pm 162.5$  MHz.

## 4 Complex Remez Algorithm

The complex Remez algorithm (also known as Lawson’s iteratively reweighted least-squares, IRLS) extends the classical Parks–McClellan algorithm to complex-valued FIR filters. At each iteration the algorithm solves a weighted least-squares problem

$$\min_{\mathbf{h}} \sum_k w_k(\ell) |\mathbf{a}(\omega_k)^\top \mathbf{h} - H_d(\omega_k)|^2,$$

and then updates the weights as  $w_k(\ell+1) = w_k(\ell) |e_k(\ell)|$ , where  $e_k(\ell)$  is the error at frequency  $\omega_k$  after iteration  $\ell$ . This reweighting progressively concentrates the design effort on the worst-case frequency points, driving the solution toward the minimax (Chebyshev) optimum.

Unlike the convex optimisation approach, no explicit passband or stopband tolerance parameters are required; the algorithm converges to the equi-ripple solution determined solely by the frequency grid and the desired response  $H_d(\omega)$ .

Figure 3 shows the full-band magnitude and phase response of the filter designed with the complex Remez algorithm. Figure 4 zooms into the passband region, and Figure 5 shows the in-band approximation error.

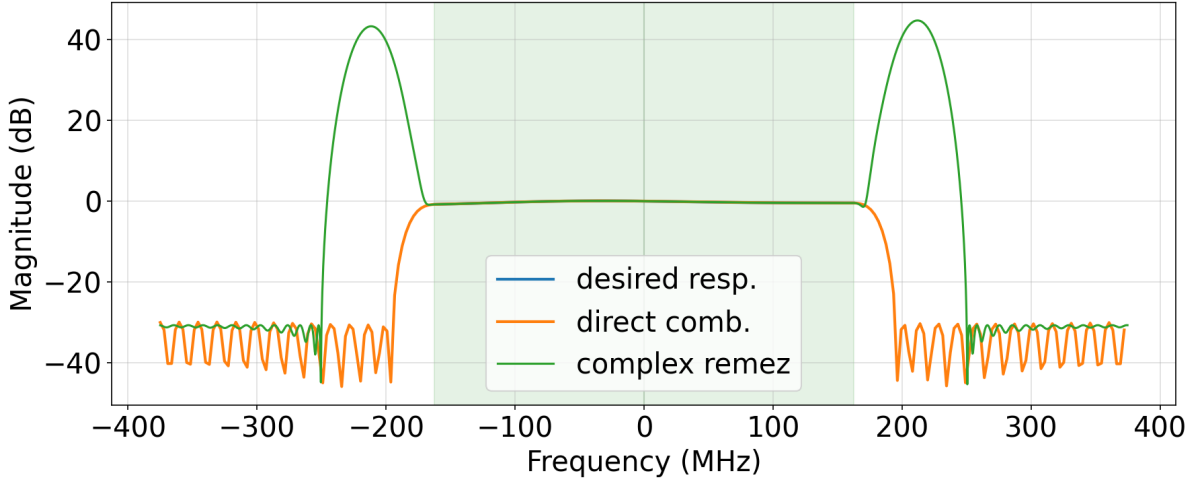


Figure 3: Full-band magnitude (top) and phase (bottom) of the complex Remez FIR filter. The green shaded region indicates the passband  $\pm 162.5$  MHz. The dashed red line is the desired response  $H_d(\omega)$ .

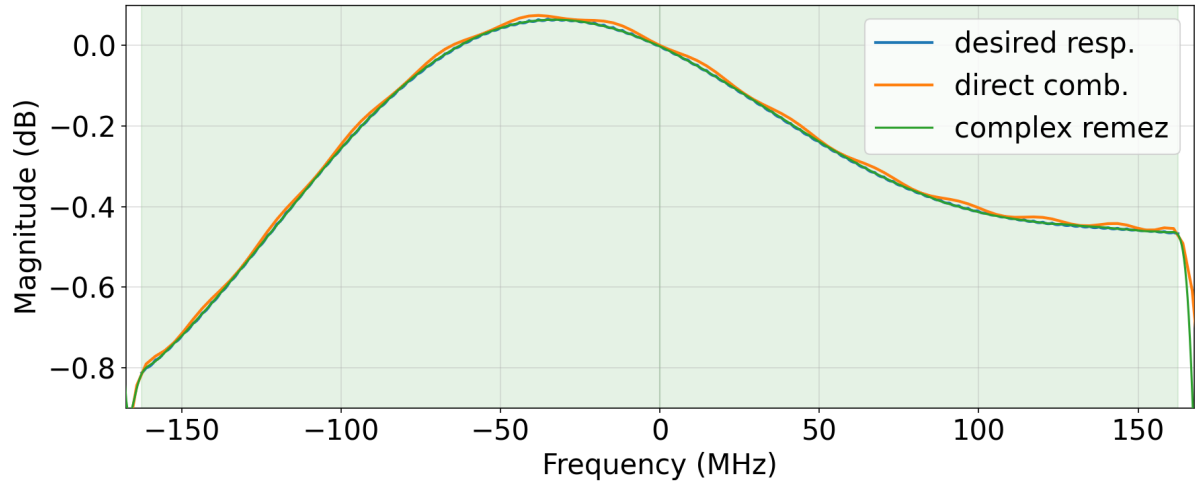


Figure 4: In-band magnitude response of the complex Remez filter zoomed to the passband region. The green shaded region indicates the passband  $\pm 162.5$  MHz.

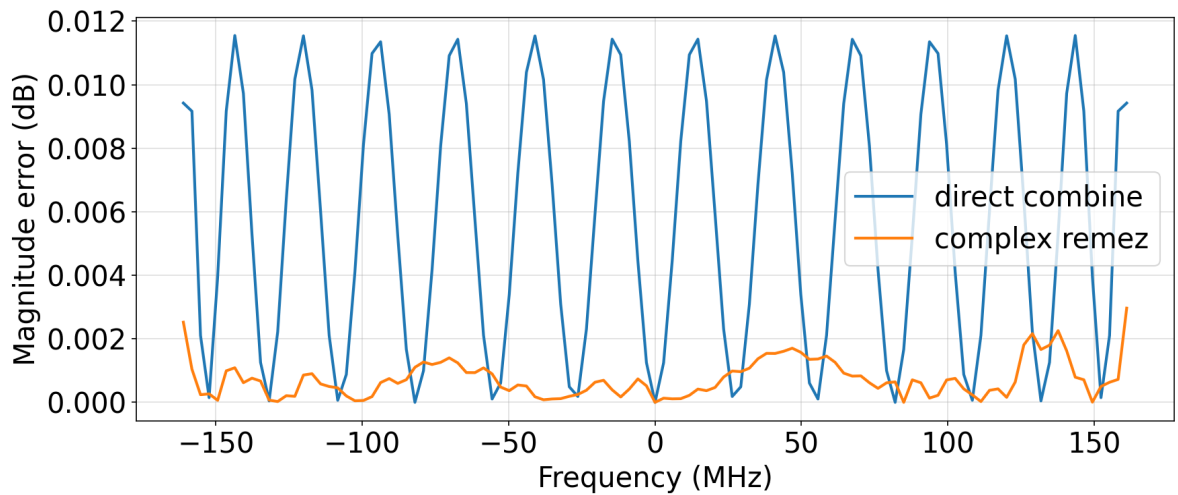


Figure 5: In-band approximation error of the complex Remez filter. The equi-ripple characteristic of the Chebyshev solution is visible across the passband.

#### 4.1 Cosine Transition-Band Constraint

Without a transition-band constraint the IRLS solver leaves the gap between the passband and stopband entirely unconstrained, which can cause the magnitude to overshoot or roll off slowly in that region. A soft constraint is introduced by appending a transition-band grid  $\mathcal{T}$  to the unified frequency grid and supplying a cosine-tapered desired magnitude  $H_d^{(\text{tb})}(\omega)$  that interpolates smoothly between the passband-edge level and the stopband target. Specifically, for a normalised parameter  $t \in [0, 1]$  sweeping from the passband edge to the stopband edge,

$$H_d^{(\text{tb})}(t) = A_{\text{stop}} + (A_{\text{pass}} - A_{\text{stop}}) \cdot \frac{1 + \cos(\pi t)}{2},$$

where  $A_{\text{pass}} = \max_{k \in \mathcal{P}} |H_d(\omega_k)|$  is the passband-edge magnitude and  $A_{\text{stop}} = \delta_s$  is the stopband target. This raised-cosine shape has zero derivative at both endpoints, giving a gentle rolloff at the passband edge and a smooth approach to the stopband level without abrupt kinks. The transition-band points enter the IRLS cost with the same band weight as the stopband ( $B_k = w_{\text{sb}}$ ), so the solver tracks the ramp target directly rather than simply suppressing energy toward zero.

Figures 6–8 show the filter designed with this cosine transition-band constraint.

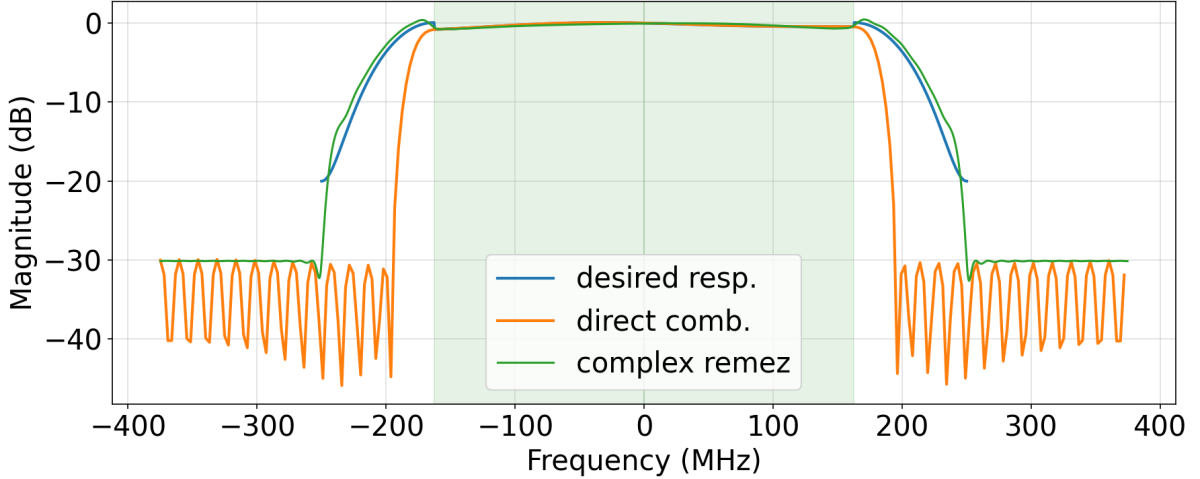


Figure 6: Full-band magnitude and phase of the complex Remez filter with cosine transition-band constraint. Compared with the unconstrained design (Figure 3), the transition region rolls off more smoothly between  $\pm 162.5$  MHz and  $\pm 250$  MHz.

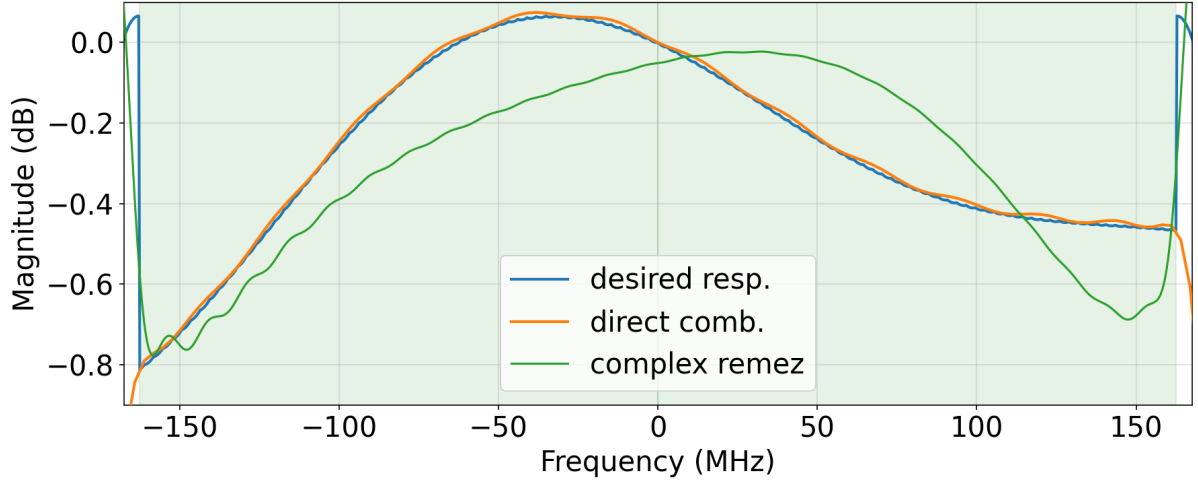


Figure 7: In-band magnitude response of the cosine-constrained Remez filter zoomed to the passband region.

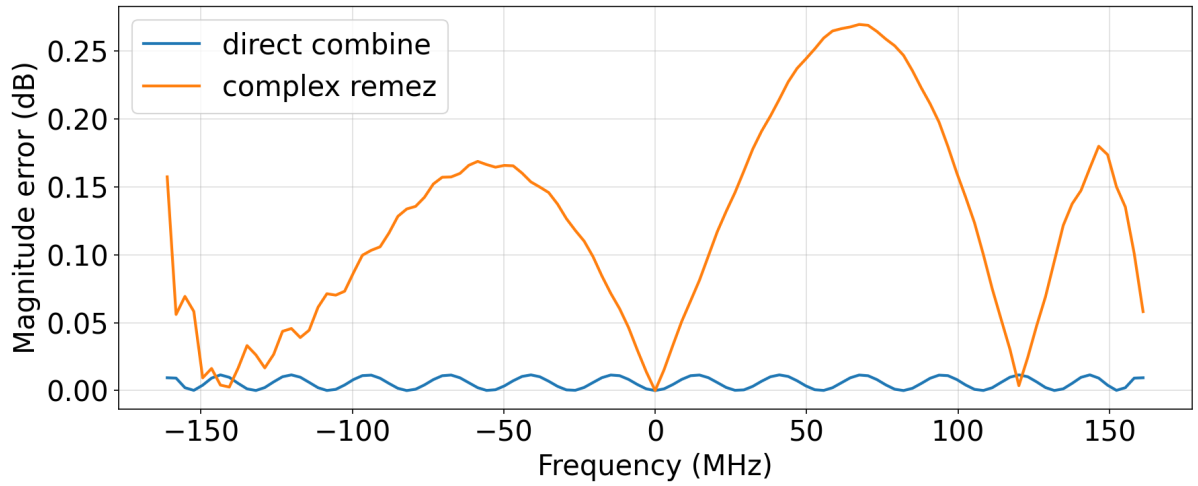


Figure 8: In-band approximation error of the cosine-constrained Remez filter. The additional transition-band degrees of freedom redistribute the equi-ripple error across the passband relative to the unconstrained result.